

Business Research Document

Data Science Capstone Project — FORGR

1. Project Overview

Project Title

FORGR — AI-Powered 360° Student Performance & Employability Analytics Platform

Domain

Education (EdTech — institutional student-performance and placement analytics), covering the academic lifecycle from enrolment through graduation and placement.

Problem Statement

A student's academic, skill, activity, and placement data is scattered across ERP systems, examination portals, placement-cell spreadsheets, sports/club records, and individual faculty notes. This fragmentation means faculty cannot identify weak students early, students don't know which skills they lack, placement teams struggle to shortlist candidates efficiently, and recruiters cannot quickly evaluate a candidate's overall profile.

Purpose of Research

To study how academic institutions currently manage and act on student-performance data, identify where the process breaks down, and determine whether a unified, predictive analytics platform can close those gaps — informing the requirement analysis and model-design phases of this capstone.

2. Domain Background

The EdTech analytics domain covers systems that track, analyze, and act on student data across a higher-education institution's lifecycle — academics, attendance, skills, activities, and placements. Institutions increasingly rely on data to identify at-risk students, measure placement readiness, and report outcomes to accreditation bodies and recruiters.

However, most B.Tech-level institutions still treat these as separate record-keeping functions rather than a connected decision-making system, so the potential of the data for early intervention and predictive insight is largely untapped. Academic marks live in the ERP, attendance is logged separately in a biometric system, skills and certifications are self-reported, and placement outcomes are tracked independently by the placement cell.

3. Current System / Existing Process

Institutions typically run several disconnected workflows rather than one system, each owned by a different department and reporting on a different cadence:

- ERP / examination portals for marks, SGPA/CGPA, and backlog records, exported per semester.
- Biometric or manual attendance registers, tracked separately from academic records.
- Spreadsheet-based placement-cell tracking of drives, interviews, and offers.
- Self-reported or ad-hoc tracking of skills, certifications, and coding-platform activity (LeetCode, HackerRank, GitHub).
- Manual faculty judgment — rather than a data-driven flag — to identify a struggling student.

4. Identified Business Pain Points

- Faculty cannot identify weak or at-risk students until results are already declared, since attendance and coursework signals are never combined into one early-warning view.
- No systematic backlog / dropout early-warning mechanism exists — institutions react after the fact rather than flagging leading indicators.
- Skill and certification data isn't tracked centrally, so gaps surface only at the interview stage.
- Placement-cell shortlisting is manual and spreadsheet-driven, with no predictive signal to rank candidates.
- Parents and recruiters have limited, inconsistent visibility into a verified student profile.

5. Impact of Pain Points

- Late intervention — at-risk students are identified only after a decline already shows in results, when it's too late to act.
- Slower, less confident placement-cell decision-making, since shortlisting relies on manual review rather than ranked, predictive signals.
- Missed placement opportunities for students whose skill gaps aren't surfaced until interviews.
- Inefficient use of placement-cell and faculty time on manual data compilation instead of intervention or mentoring.
- Weaker institutional reporting to recruiters and parents, affecting trust and placement outcomes.

6. Data Availability & Sources

Both structured and semi-structured data are required, drawn from institutional and public sources:

- Academic: subject marks, SGPA/CGPA, backlogs — from ERP / examination portal exports (structured).
- Attendance: subject-wise attendance % — from ERP or biometric attendance systems (structured).
- Skills: self/faculty ratings, coding-platform statistics — from student self-entry and LeetCode/HackerRank/GitHub exports (semi-structured).
- Certifications: certificate name, issuer, date — from student upload with issuer verification (semi-structured).
- Activities & soft skills: sports/cultural records, club roles, faculty evaluation forms (semi-structured).
- Placement: resume/ATS score, interview outcomes, offers — from placement-cell records and resume upload (structured).
- Public datasets (e.g., Kaggle student-performance datasets) to supplement prototyping where institutional data is unavailable.

7. Existing Solutions / Competitor Analysis

Institutions today rely on generic or point-solution tools rather than a purpose-built, predictive student-analytics platform:

- Institutional ERPs (e.g., Fedena-style systems) — handle academic records and attendance, but offer no predictive risk modeling or unified skill/placement profile.
- General-purpose BI tools (Power BI, Google Analytics-style dashboards) — can visualize institutional data if fed into them, but require manual data preparation and no built-in employability scoring.
- Placement-management or campus-hiring platforms (e.g., campus modules on Naukri/LinkedIn-style job portals) — support drive and application management, but don't ingest academic or skill data for predictive shortlisting.

Limitations across the board: no single 360° student profile, no backlog/dropout risk prediction, and no role-specific dashboards for student, faculty, parent, and recruiter needs simultaneously.

8. Identified Gaps

- No unified 360° profile spanning academics, skills, activities, and placement in one place.
- No predictive early-warning system for backlog, dropout, or attendance risk.
- No systematic, comparable employability scoring for placement shortlisting.
- Limited automation — most current processes still depend on manual compilation and judgment.
- No role-based personalization tailored to students, faculty, placement cell, parents, and recruiters.

9. Target Users / Stakeholders

FORGR is designed around six distinct stakeholder roles, each with a different view into the same underlying student data:

User Role	Responsibility
Student	Track own academic, skill, and activity growth; choose which skills/certifications to pursue
Faculty / Mentor	Identify and intervene early with at-risk students (backlog, attendance, dropout)
Placement Cell	Shortlist and rank candidates for placement drives
Parent	Monitor ward's academic and behavioural progress
Recruiter	Evaluate consolidated, verified candidate profiles quickly
Institution Admin / HOD	Oversee data-sharing approvals and institution-wide adoption

10. Proposed Data Science Approach

FORGR combines descriptive profile aggregation with predictive analytics:

- Weighted scoring model to compute a 0–100 Employability Score from academic, skill, coding, project, certification, communication, leadership, and attendance features.
- Classification models (Logistic Regression, Random Forest, XGBoost) to predict backlog, dropout, and low-attendance risk.
- Classification for placement-probability prediction (low/medium/high) from the aggregated feature set.
- K-Means clustering to group students into cohorts for comparative analysis.
- All models recompute whenever underlying source data is updated, so scores and risk flags stay current.

11. Expected Business Impact

- Faculty can flag and intervene with at-risk students at least one full grading cycle before the outcome would otherwise surface.
- Placement-cell shortlisting time drops from manual hours to a filtered, ranked list in under 5 minutes.
- Higher faculty adoption of risk dashboards and higher student profile-completion rates.
- More consistent, comparable candidate evaluation for recruiters, improving placement outcomes.
- A reusable, scalable data model that grows from a single pilot cohort to the full institution without redesign.

12. Research Conclusion

- Student data across institutions is fragmented across ERP, examination, placement, and activity records, with no unified view.
- This fragmentation delays intervention for at-risk students and slows placement-cell decision-making.
- Existing ERP and BI tools address parts of the problem but none combine a 360° profile with predictive risk and employability scoring.
- A weighted employability-scoring model paired with classification-based risk prediction can close this gap.
- FORGR's proposed platform, if validated against real cohort data, can meaningfully improve early intervention, placement-shortlisting efficiency, and stakeholder visibility.